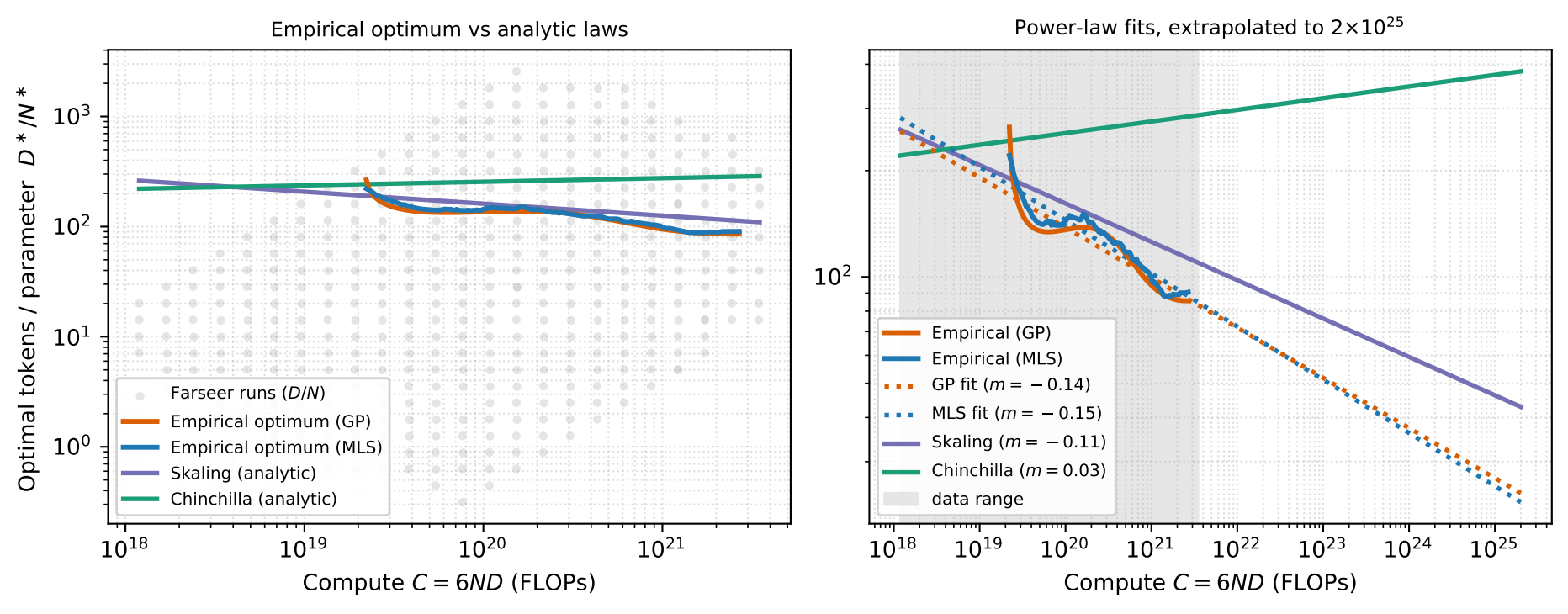
Empirical optimum vs analytic laws
Optimal tokens / parameter $D^*/N^*$
$10^3$
$10^2$
$10^1$
$10^0$
Farseer runs ($D/N$)
Empirical optimum (GP)
Empirical optimum (MLS)
Skaling (analytic)
Chinchilla (analytic)
$10^{18}$
$10^{19}$
$10^{20}$
$10^{21}$
Compute $C = 6ND$ (FLOPs)
Power-law fits, extrapolated to $2\times10^{25}$
$10^2$
Empirical (GP)
Empirical (MLS)
GP fit ($m = -0.14$)
MLS fit ($m = -0.15$)
Skaling ($m = -0.11$)
Chinchilla ($m = 0.03$)
data range
$10^{18}$
$10^{19}$
$10^{20}$
$10^{21}$
$10^{22}$
$10^{23}$
$10^{24}$
$10^{25}$
Compute $C = 6ND$ (FLOPs)